Vox Populi
Exploring the Strengths and Limits of Collective Decision-Making *

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"In general, a law which has not been voted unanimously involves subjecting men to an opinion which is not their own, or to a decision they believe contrary to their interest. It follows that a very great probability of the truth of this decision is the only reasonable and just grounds according to which one can demand such submission."

"The truth belongs to those who seek it, not to those who claim to own it."

- Marie-Jean-Antoine-Nicolas de Caritat, Marquis de Condorcet (September 17, 1743–March 28, 1794)

Chances are that if you are reading this in early 2024, there is an election looming in the horizon and news about it are occupying a significant portion of your daily news feed, whether you like it or not. Predictions about the results will slowly start emerging as we draw closer, and will be fiercely debated over the next few days. Elections are perhaps one of those situations where prediction accuracy trumps interpretability—and the stakes are pushed to the extremes — when forecasters get it right, they are bestowed with too much credit, and when they are off the target, the credibility of their entire trade is under mass scrutiny.

I am not a psephologist or a political analyst by any stretch of imagination, although I must admit that I have enjoyed the vicarious glory when Nate Silver rose to stardom and on rare occasions, I might have even used his oft-quoted remark on the difference between data science and statistics [1]. For this article, I will stick to a more fundamental, epistemic question that has perplexed thinkers for a really long time: is the *vox populi* always correct? To make things simpler, assuming there is an objective truth, can a group of individuals arrive at the correct decision by majority voting? If yes, under which conditions, and if not, will it be a complete disaster?

*This is an article for my column 'Datta's Data Dives' for the IISA Newsletter.

[1] When asked about the difference between statistics and data science during JSM 2013, Silver famously replied, "I think data-scientist is a sexed up term for a statistician".

The history or philosophy of science is rife with attempts to formalize the notion of 'wisdom of the crowd' from the ancient ages, and Aristotle is often credited by historians as one of the first people to recognize the power of information aggregation. The reader, a student of statistics, is probably more familiar with the 'wisdom of crowd' estimation by Sir Francis Galton, published as an article entitled *'Vox Populi'* in the March edition of Nature [Galton, 1949]. If not, here is a shortened version: Galton [1949] describes a weight-judging competition in Plymouth where a fattened ox is selected, and competitors, each paying six pennies, guess the weight after the ox has been slaughtered and dressed, where the most successful guesses would receive a prize. Galton commented that the six penny entry-fee would prevent people from practical joking, and the prize money would encourage putting the best bet, and the competitors included butchers and farmers, experts in the matter. After discarding 13 of the 800 guesses, Galton observed that the median guess (his preferred measure), 1208 pounds, was within 1% of the true weight, 1197 pounds, and the mean would coincide with the actual weight, producing a perfect estimate [Wallis, 2014]. Galton, at the end of the article, remarked that, "This result is, I think, more creditable to the trustworthiness of a democratic judgment than might have been expected." Indeed, 'wisdom of crowds' is pervasive in almost all human endeavors wherever decision-making and uncertainty work together: when the truth is singular, and the errors are 'random' (and not systematic), averaging must improve accuracy. I would defer interested readers to the popular book, "The Wisdom of the Crowds" [Surowiecki, 2005] for a broader perspective.

But, the first *rigorous* mathematical foundation for 'wisdom of the crowd' dates back to the Age of Enlightenment, just before the French Revolution had swept France, to a French philosopher and mathematician, *Marie Jean Antoine Nicolas de Caritat, Marquis de Condorcet*, or simply Condorcet. Often called the last of *philosophes*, or the last witness of the Age of Enlightenment, Condorcet was a fascinating and tragic hero of his times, and a proponent of human rights and equality for all, making him exceptional even amongst enlightened thinkers. Among other things, he advocated for educational reforms, representative government, abolition of colonial slavery and equal rights for women, especially women suffrage and argued for equality of all races, and formulated human progress via indefinite perfectibility of humankind and society. Condorcet also pioneered statistical and probabilistic thinking into social sciences, via 'social arithmetic', and apply mathematical calculations to financial planning, jury decision-making, public health and other fields.

Condorcet

Coming back to our main story, it was this Nicolas de Condorcet who, 164 years before Galton, laid out the first precise mathematical calculations for the majority voting rule, or the

'wisdom of the crowd', via the famous Condorcet Jury Theorem, or CJT [Condorcet, 1785]. I will state the simplest version here and point interested readers to Goodin and Spiekermann [2018] for a book-length treatment. Assume there exists an objective truth or a true state of the world and an odd n number of voters, each with probability p_c of voting for the correct alternative, arrive at the collective decision by a majority rule. Then, the classical version of CJT asserts two things. First, in a group of 'independent', 'competent', and 'sincere' voters, where each voter is more likely to be right than wrong ($p_c > 1/2$), the majority vote is more likely to be accurate than any individual vote. Second, as the number of such voters approaches infinity, the probability that the majority is correct approaches 1. In familiar notations, if P_n denotes the probability that the majority rule arrives at the correct outcome, it's easy to derive that,

$$P_n = \sum_{k=\frac{n+1}{2}}^{n} \binom{n}{k} p_c^k (1-p_c)^{n-k} \approx 1 - \Phi\left(\frac{n/2 - np_c}{\sqrt{np_c(1-p_c)}}\right).$$

Thus, the CJT asserts that for every n and every $p_c > 1/2$ (a) $P_{n+2} > P_n$ and (b) $P_n \to 1$, as $n \to \infty$, as the upper half of Figure 1 shows.

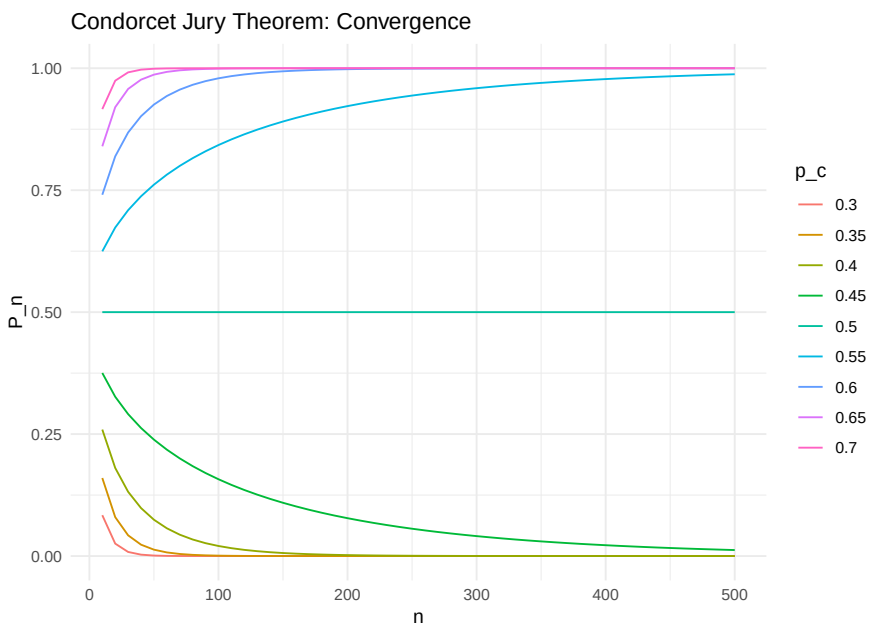

Figure 1: Convergence of P_n as n grows for different levels of individual competence p_c

This remarkable result from the very early days of probability was largely lost to history [Black et al., 1958], along with Condorcet's other famous paradox, where he showed that majority voting could lead to cyclic preferences, even when the individual preferences are not cyclic, *i.e.*, A is preferred to B and B is preferred to C, and yet C is preferred to A, by the majority.

Of course, the idea of information aggregation or majority voting has applications in our

own field, for example, ensemble learners like random forest are based on the idea that aggregating predictions from multiple models can improve prediction accuracy or classification performance, when they are better than random and sufficiently decorrelated. I will present here a simple result for such a situation, based on an exercise from Vershynin [2018][2]:

Consider a randomized algorithm for decision-making (e.g., determining if a number is prime) that gives correct answers with probability $\frac{1}{2} + \delta$*, where* $\delta > 0$ *(i.e., assuming 'competence'). By running the algorithm* n *times and taking the majority vote, the probability of obtaining the correct answer would be at least* $1 - \epsilon$ *for any* $\epsilon > 0$*, as long as,* $n \geq \frac{1}{2\delta^2} \ln(\frac{1}{\epsilon})$*.*

But, is the majority always correct, or, is CJT always applicable in *the real world*? Recall that CJT only requires each voter to be barely better than random, that they arrive at their judgment independently of each other, and they 'all express their opinion in good faith'. These assumptions can fail. Indeed, experts have raised objections to all three of them: competence, independence and sincerity, and examined how they can be violated and also to what extent those objections can be forestalled [see Goodin and Spiekermann, 2018]. As the lower half of Figure 1 suggests, if each voter is more likely to be wrong than right, *i.e.,* their 'individual competence' p_c is less than $1/2$, more voters will in fact make things worse and the optimal jury shall consist of a single expert.

Perhaps more skepticism is warranted for the assumption of 'independence' which can fail if voters have 'systematic' biases or uncritically follow the same propaganda or are subjected to the same misinformation or have the same prejudices. An interesting paper along these lines is Hill and Garner [2021] who question applicability of CJT to the case of virtue signalling, as our strong conformist tendencies make us less independent. Condorcet himself expressed concerns that the masses suffer from 'great ignorance with many prejudices', and unfortunately it still rings true after almost 250 years.

Condorcet, a perfect embodiment of the era of Enlightenment, spent his last days in hiding, while writing his greatest work, the *Esquisse*. *Esquisse* presents the idea of continuous human progress towards ultimate perfection, envisioning stages of enlightenment and equality culminating in a future epoch marked by the destruction of inequality between nations and classes. On the morning of March 29, 1794, which is exactly 230 years back to this date, Nicolas de Condorcet was found dead in his prison cell, cause of death unknown.

This article has now grown lengthier than it was supposed to, and I must end it here. As we look towards an uncertain future with equal parts hope and trepidation, I will leave the readers with one of my favorite quotes by Condorcet, translated by Landes [2009]:

"*I believe humankind is infinitely perfectible, and that it should thus devote itself to achieving peace, liberty and equality, whose term is impossible to fix. I also believe that this progress must be the work of reason, fortified by meditation, supported by experience.*" — Condorcet, under the pseudonym "Un Vieux Bramine".

[2] As Vershynin remarks, the proof is immediate by applying Hoeffding's inequality.

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